ARCHITECTURE AND DECISION RECORD

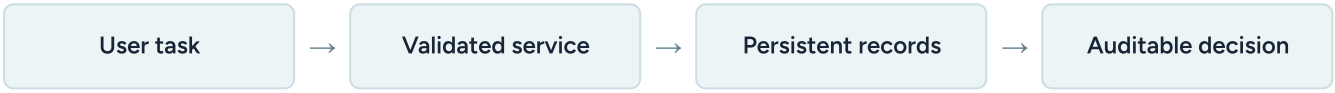
A supervised environmental monitoring and actuator prototype

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ESP32 MQTT Node.js Sensor calibration

Problem and contribution

This original design case study explores how to monitor greenhouse conditions and control a small irrigation prototype within explicit calibration and actuator safety boundaries. I authored the requirements breakdown, architecture boundaries, data relationships, failure scenarios, and review rationale presented here. The design is a fictional presentation example, with no claim of a production deployment or measured client result.



Architecture decisions and tradeoffs

- A local controller stops the pump even if the broker, dashboard, or server disappears.
- Treat MQTT delivery as potentially duplicated or delayed; record both measured_at and received_at.
- Reject closed-loop control when readings are stale or out of calibrated range; surface the reason to the operator.

Core records

Record	Responsibility
devices	Each registered controller has multiple sensors and publishes bounded telemetry.
sensors	A sensor belongs to one controller and references the current calibration record.
calibrations	Calibration history is retained instead of overwritten with the latest adjustment.
measurements	A device/sensor/sequence constraint prevents duplicate MQTT delivery from duplicating observations.

The service layer owns authorization and business constraints. Interface validation improves the experience but cannot replace server checks. Event records preserve attributable decisions and their timing without logging unnecessary private content.

VERIFICATION AND REVIEW PRACTICE

Failure cases, evidence, and limitations

Proposed verification matrix

Scenario	Method	Expected behavior
Calibration repeatability	Compare multiple probe readings at dry, intermediate, and wet reference conditions.	Report error and spread, not a single unqualified accuracy percentage.
Duplicate delivery	Replay identical packets and delayed measurements out of order.	Store one logical observation per sequence and preserve measurement time.
Fail-safe pump control	Disconnect the broker and restart the dashboard during a watering cycle.	The local maximum runtime still stops the pump and the next connection reports the event.
Stale or failed sensor	Freeze readings or inject an impossible value while an irrigation rule is active.	The rule is disabled safely and the operator sees an actionable fault message.

Technical review

Record measured and received timestamps separately, deduplicate sequence numbers, and enforce pump runtime locally. Disconnect the broker during a test to confirm that a lost network cannot leave an actuator running.

How to evaluate the design

Report absolute sensor error against the selected reference, repeatability, delivery delay, duplicate rejection rate, command acknowledgement time, and maximum observed pump runtime. Crop growth and yield are outside this bench study; no agronomic improvement is claimed.

The matrix specifies intended tests, not completed test results. A reviewer should capture the environment, initial records, inputs, expected state, observed state, and any unresolved defect before interpreting the outcome. Repeated runs should use separate unique request keys where relevant and preserve failures for analysis.

Scope limitations

No unattended agricultural deployment, mains wiring instructions, fertilizer dosing, crop-yield prediction, or claim of water savings before a controlled study.

Review the separation between server-side visibility and controller-side safety; a successful dashboard update must not be mistaken for verified actuator behavior.

> **Authorship and reuse**
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> Prepared by Marco Santiago as an original presentation work sample. All narrative, schemas, and test scenarios are original illustrative material. There are no external client documents, employment certificates, real personal records, or fabricated citations in this case study.